

Lab -1

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EXPERIMENT 2

Objective / Aim

1. To get familiarised with Arduino and Tinkercad simulation software.
2. To turn ON and turn OFF a LED continuously in loop using Arduino.

Components used

- a LED (red coloured)
- breadboard
- Arduino Uno
- Resistor
- Two wire (preferably one black and the other red)
- USB cable

Hypothesis

We can write code like

1. `digitalWrite(pin number, high/low)`

which will show light when high and no light when low.

So basically, it gives binary input 1 which is electricity at high and binary input 0 which is no electricity at low.

Further, we can delay the time after which a low or a high comes. We use “`delay(time in milliseconds)`” to do the same.

Procedure

- Step 1 - Connect a (red)wire to one of vertical strip's hole.
- Step 2 - Connect a resistor to one of the same vertical strip's holes
- Step 3 – Connect that (red)wire other end to one of the digital pin. (mine is 8)
- Step 4 – Connect the black wire with one of the adjacent vertical strip to the strip where the other end of resistor ended.

- Step 5 – Connect the other end of black wire to the ground of Arduino Uno
- Step 6 – Connect the LED's cathode in the same vertical strip's hole as the black wire and the anode end in the same vertical strip as the resistor.
- Step 7- Connect the USB to your laptop.
- Step 8- Write the code on Arduino and run it.

Observation

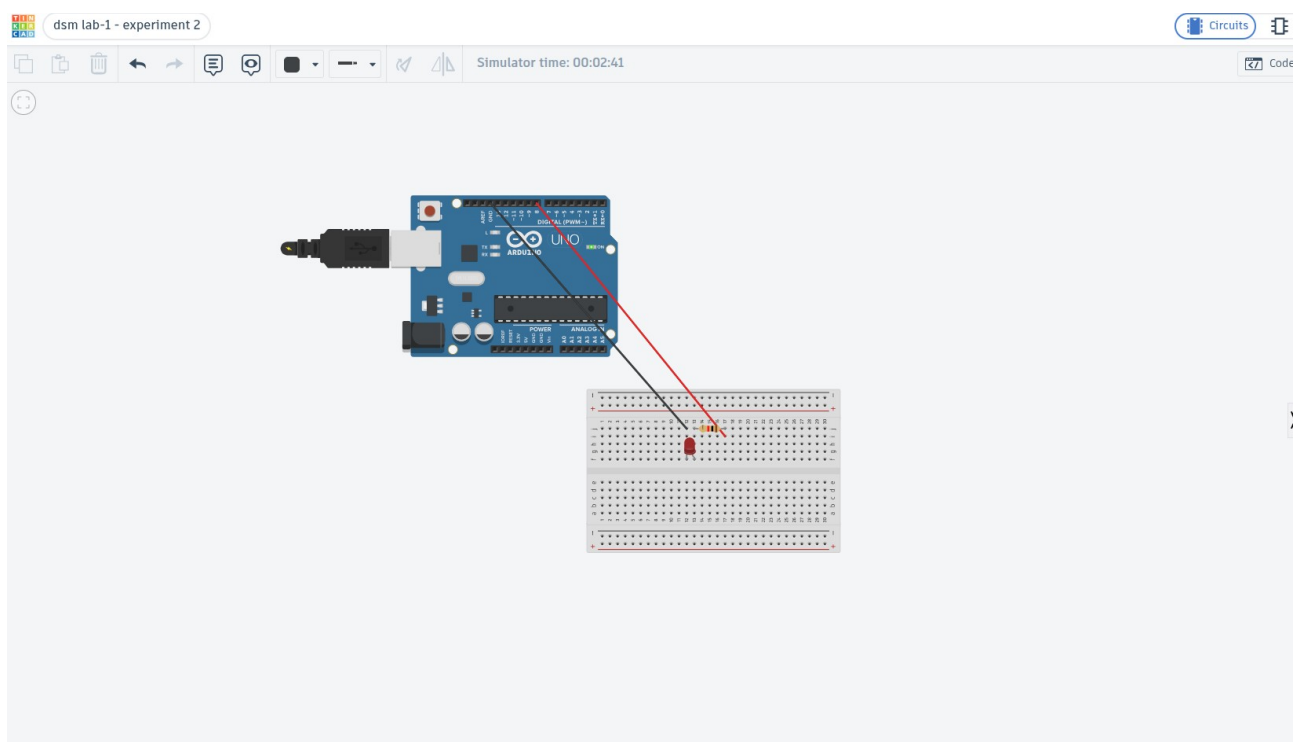
1. The LED bulb glows on and off.
2. The LED glows on and off after every 1000 milliseconds.

When we run the programme written in Arduino while connecting it to our laptop the bulb starts glowing for 1000 milliseconds and then is switched off for 1000 milliseconds .This happens continuously until we stop the programme or disconnect the USB.

Simulation Link (TinkerCAD)

https://www.tinkercad.com/things/4Z6hqdpbFsv/editel?sharecode=Po5B-YPcsZ628Tr50R5EdDT_LZLZ0Z3aNMF62q--90

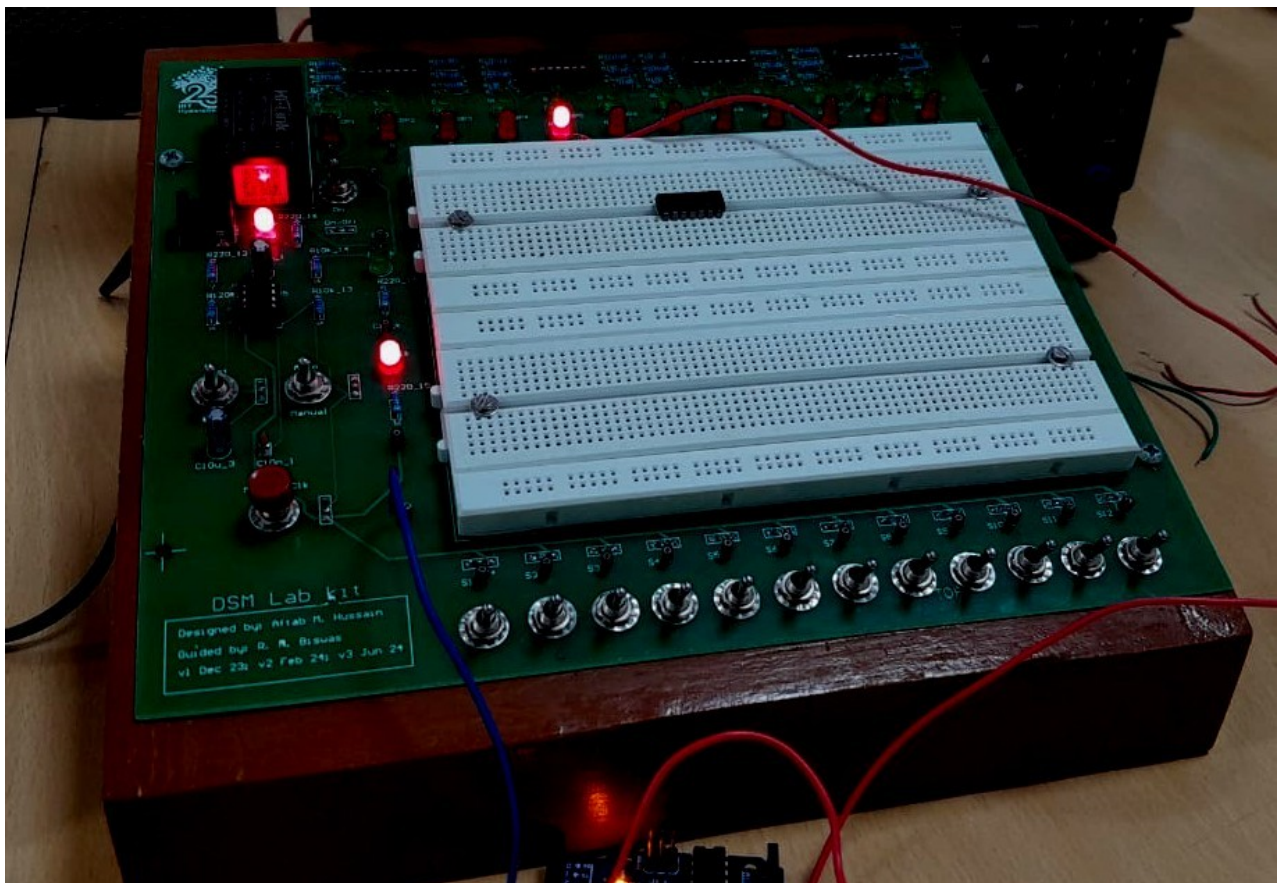
TinkerCAD Screenshot



Arduino code

```
1 // C++ code
2 void setup()
3 {
4   pinMode(8, OUTPUT);
5 }
6
7 void loop()
8 {
9   digitalWrite(8, HIGH);
10  delay(1000); // Wait for 1000 millisecond(s)
11  digitalWrite(8, LOW);
12  delay(1000); // Wait for 1000 millisecond(s)
13 }
```

Lab Photograph



Conclusion

The light of LED switches on and off at every interval of 1000milliseconds according to our code in Arduino. So, the LED looks like it is blinking.